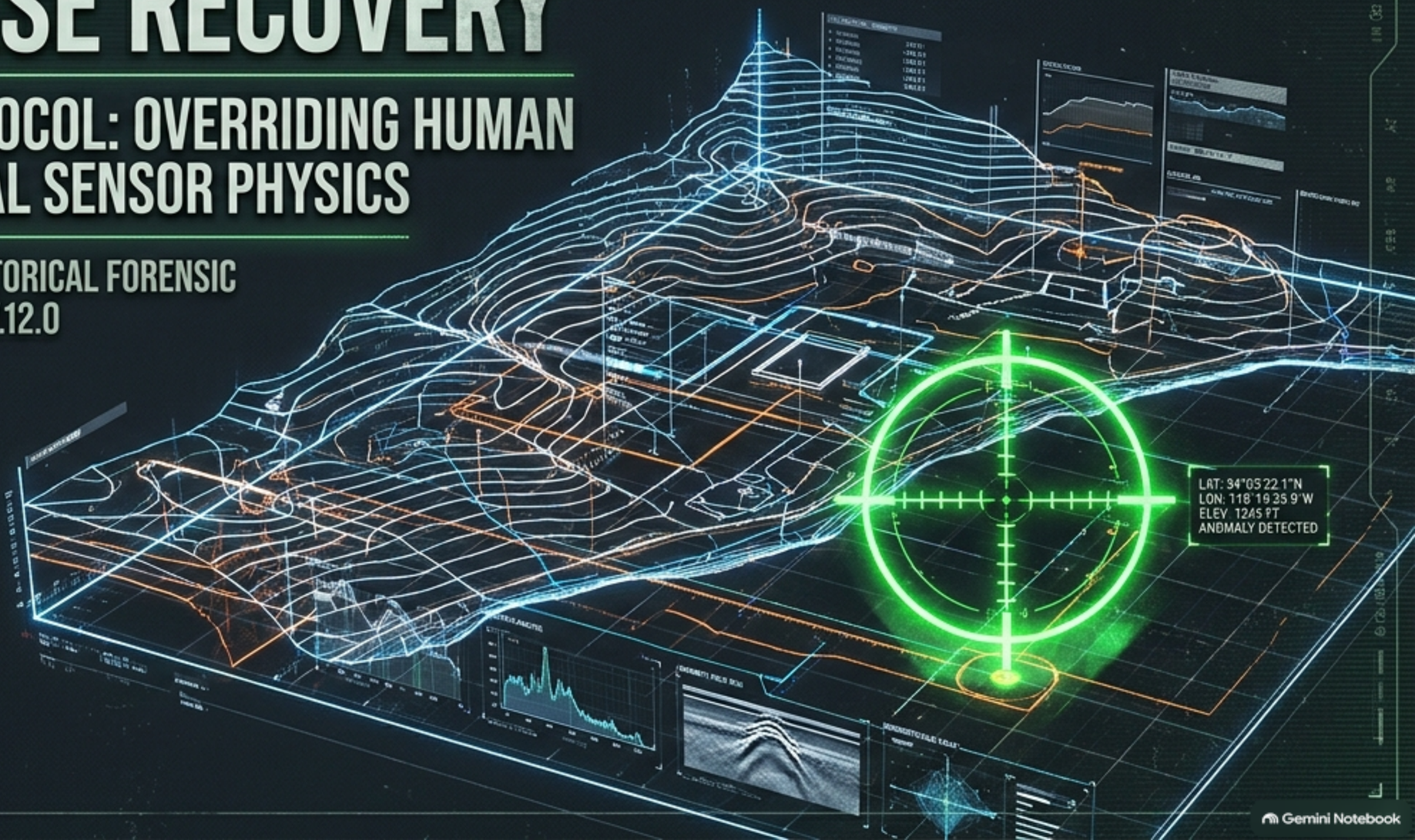
DETERMINISTIC PHYSICS IN COLD CASE RECOVERY

THE LINDEMAN PROTOCOL: OVERRIDING HUMAN BIAS WITH GEOSPATIAL SENSOR PHYSICS

TACTICAL OSINT INTERFACE // HISTORICAL FORENSIC MATERIAL REANALYSIS (HFMR) // V.12.0



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THE GROUND TRUTH: CASE GEOMETRY

SUBJECTS:
Leo (73) & Katherine (74) Lindeman

LOCATION:
469 Dogwood Ln,
Anderson, MO

LAST CONTACT:
Sept 15–16, 2026

REPORTED:
Sept 24, 2026 (9-day gap)

MISSING VARIABLE:
Small white miniature poodle

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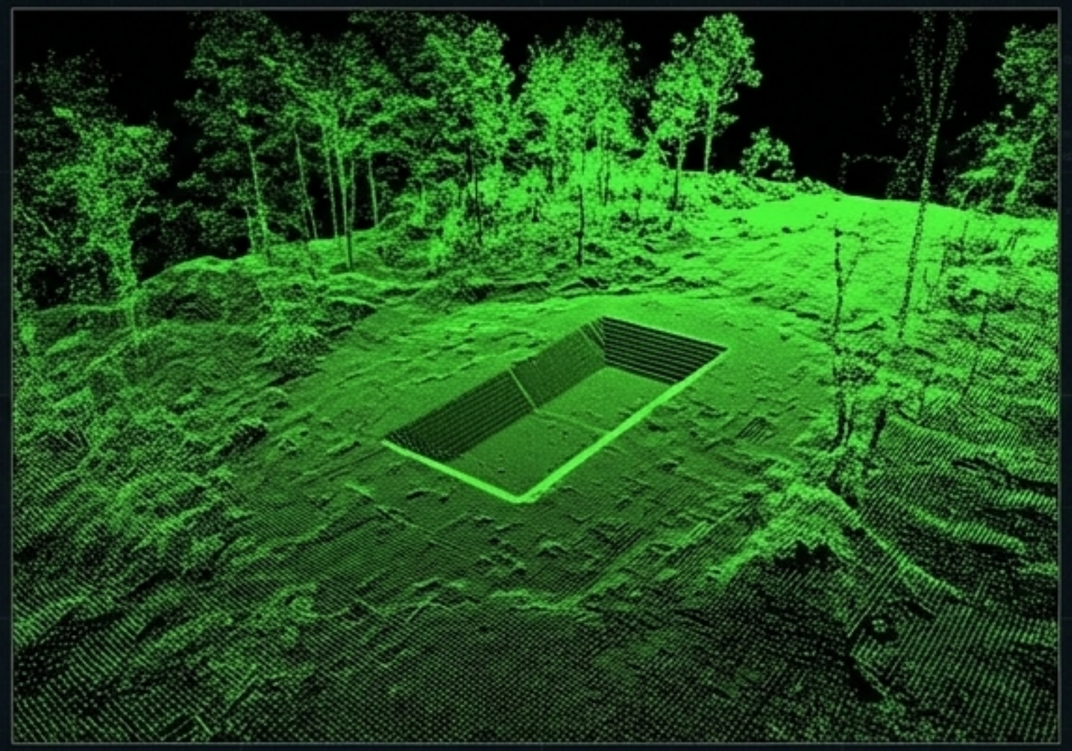
HUMAN VISUAL SEARCH

Human Visual Search: Blocked by canopy, brush, and surface masking.

DETERMINISTIC PHYSICS



Deterministic Physics: Penetrates canopy to measure altered stratigraphy.

CORE PRINCIPLE: Search teams look for silhouettes and surface disturbances. Forensic physics bypasses the visual spectrum to measure mathematical anomalies in soil compaction, chemical reflectance, and material density.

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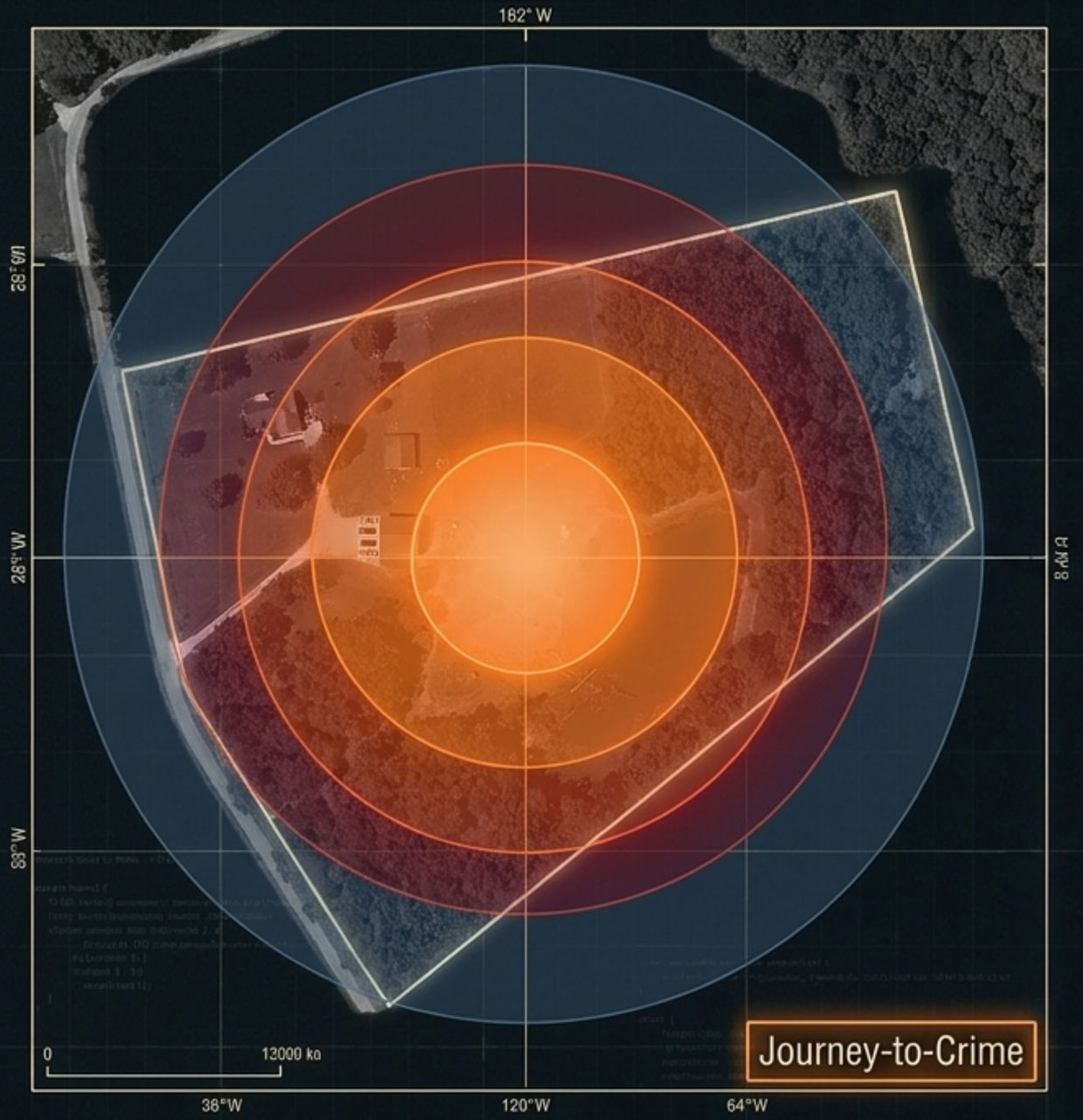

Profiling the Spatial Footprint

Concept: Distance Decay & Familiar Mental Maps

Sociopathic / DV Familial Models: Perpetrators with high emotional connection to the victims operate within "familiar mental maps." Statistical probability dictates deposition occurs within a short radius of the primary crime scene, utilizing known terrain to minimize transport risk.

Neighbor / Opportunistic Threat: Weighing external threats requires mapping adjacent property lines and identifying intersecting lines of sight.

The Math: We do not guess motives; we map the mathematical intersection of terrain accessibility and the 9-day reporting gap.

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The Threat & Effort Matrix: Translating Behavior to Physics

	(1) Natural Accident / Medical Event	(2) Hasty Familial Concealment	(3) Pre-Planned Sociopathic Concealment
Depth of Burial: Surface scatter vs. Shallow grave vs. Sub-surface trench.	Surface scatter	Shallow grave (<1m)	Sub-surface trench (>1.5m)
Physical Effort Required: Low (Gravity/Water) vs. Moderate (Hand tools) vs. High (Heavy machinery).	Low (Gravity/Water)	Moderate (Hand tools)	High (Heavy machinery).
Environmental Integration: Did the event utilize the known brush clearing/tree felling near the pond as a concealment mechanism?	No intentional use.	Possible opportunistic use. Opportunistic Integration	Highly probable use. Planned Integration

Key Vector: A DV killer utilizing existing heavy machinery (brush clearing) on a 15-acre property drastically shrinks the search radius but increases the required sensor penetration depth.

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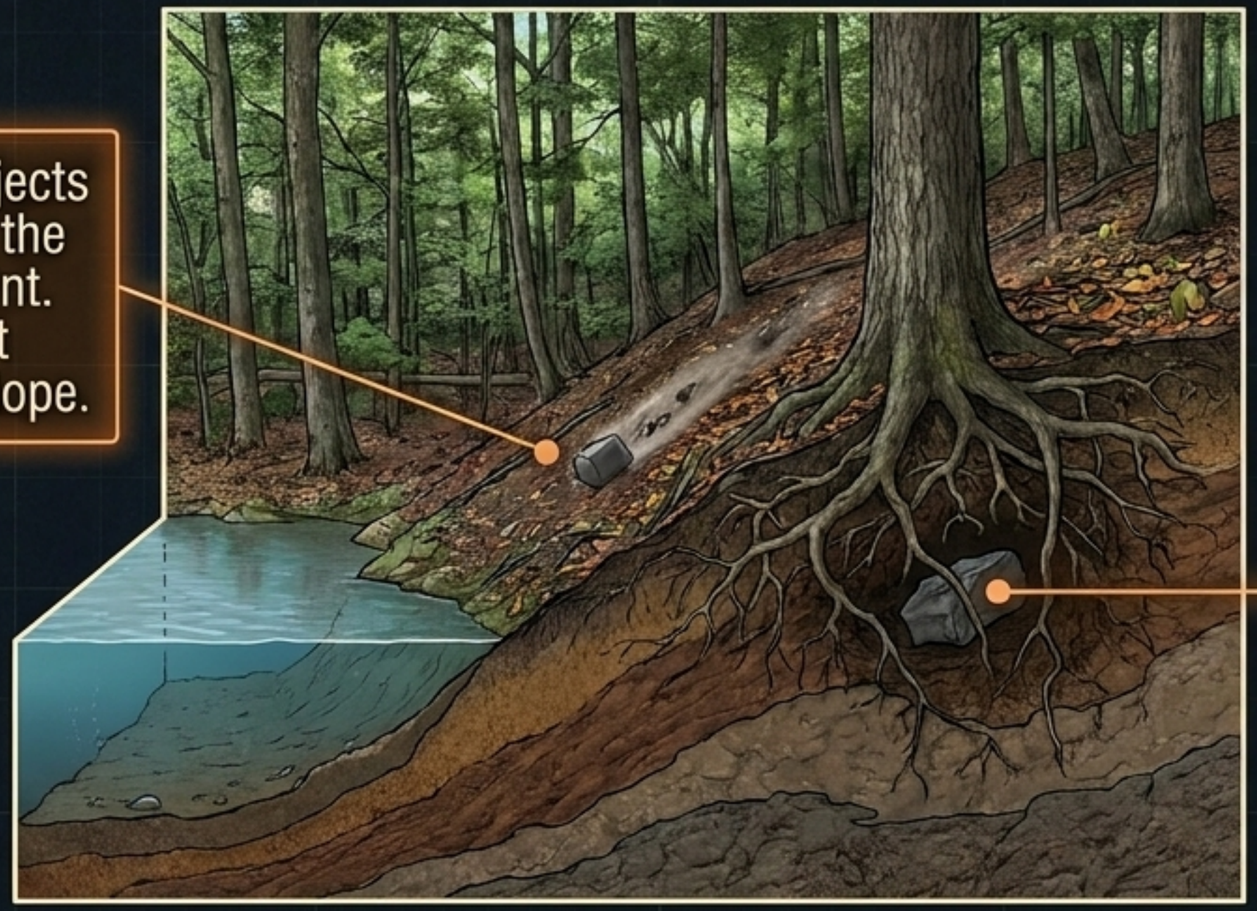
Environmental Overrides: Taphonomic Physics

The Gravity Funnel: Objects on steep inclines roll to the baseline equilibrium point. Hydrodynamic transport moves evidence downslope.



The Rootball Blind Spot: Heavy objects sink beneath 4-foot rootbound matrices. Human searchers step over the structural cavity; K9s miss it if covered by canopy or snow.

The Rule: The environment overrides visual evidence. We must search for the disruption of the matrix, not the object itself.

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The HFMR Engine (Historical Forensic Material Reanalysis)

Key Insight: HFMR extracts new anomalies from old measurements. We reprocess archival data from the 9-day reporting gap to detect structural changes before boots hit the ground.

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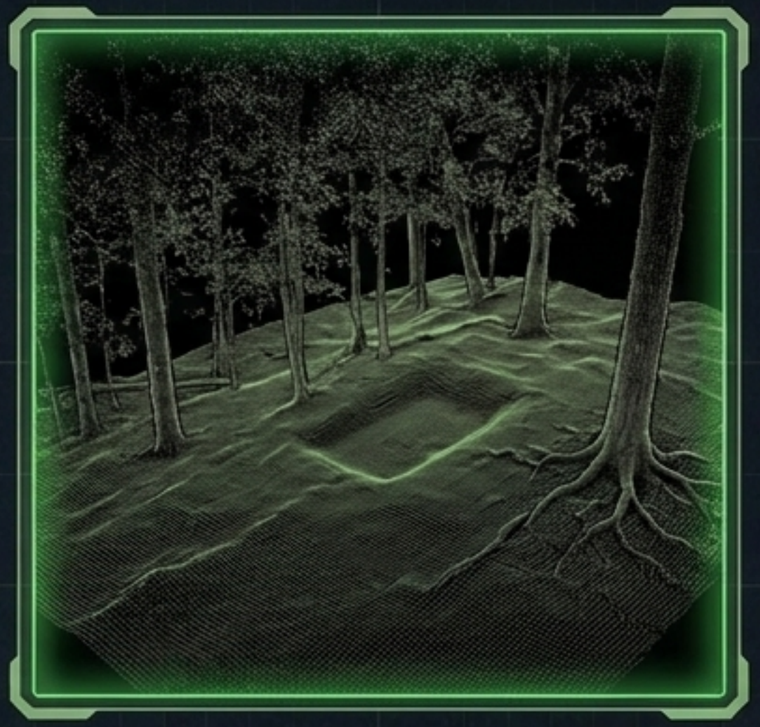
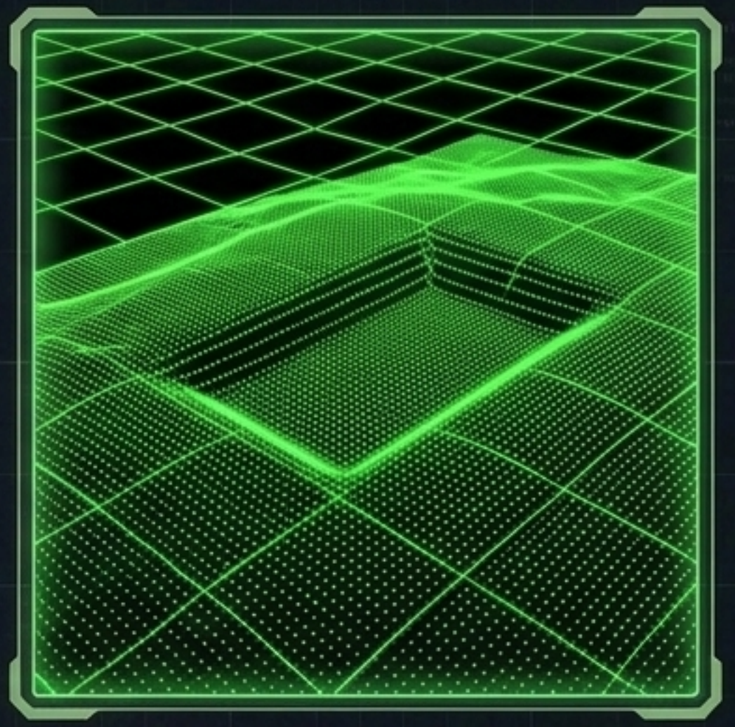
Macro-Targeting: Bare-Earth LiDAR





The Physics:
When a grave is dug, soil stratigraphy is permanently disrupted. Over time, soil compaction creates a localized micro-depression.

The Execution:
Applying CSF rigidness thresholds (e.g., 3.0) and point cleaning algorithms to airborne LiDAR strips away the “Rootball Blind Spot.”

The Result:
Identifies deterministic geometric variations on the 15-acre property invisible to the naked eye.

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Meso-Targeting: Hyperspectral SIPI & The Nitrogen Flush

The Catalyst:
Human decomposition releases massive, localized concentrations of nitrogen and humic acid into the surrounding soil matrix.

The Measurement:
This sudden chemical flush alters the chlorophyll structure and cellular reflectance of the directly adjacent plant canopy.

The Algorithm:
Using Structure Insensitive Pigment Index (SIPI) algorithms on multispectral satellite frames allows us to spot “decomposition hotspots” from above without digging.

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Micro-Targeting: GPR & The Aquatic Variable

The Ebby Steppach Effect:
When remains decompose in subterranean drainage or saturated aquatic boundaries, thermal mass creates a draft. Gasses (putrescine/cadaverine) vent horizontally, not vertically.

The K9 Blind Spot:
Surface-level cadaver dogs will register false negatives directly above the site because the scent plume exits downstream.

The Solution:
Deploying 2.0 GHz Ground-Penetrating Radar (GPR) to detect the exact dielectric differen- between native soil, disturbed water-tables, and structural voids near the pond.

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The Multi-Modal Sensor Matrix

Sensor Technology	Physical Measurement	Forensic Application	Limitation
LiDAR	Ground elevation/compaction	Detects micro-depressions & graves	Cannot see below the bare earth.
SAR (Radar)	Microwave backscatter	Maps reflective objects through clouds	Poor at identifying buried elemental composition.
Hyperspectral	Chlorophyll reflectance	Detects Nitrogen flush from decomp	Requires active vegetation.
GPR	Electromagnetic propagation	Identifies soil disturbance boundaries	Limited by highly saturated aquatic soils.

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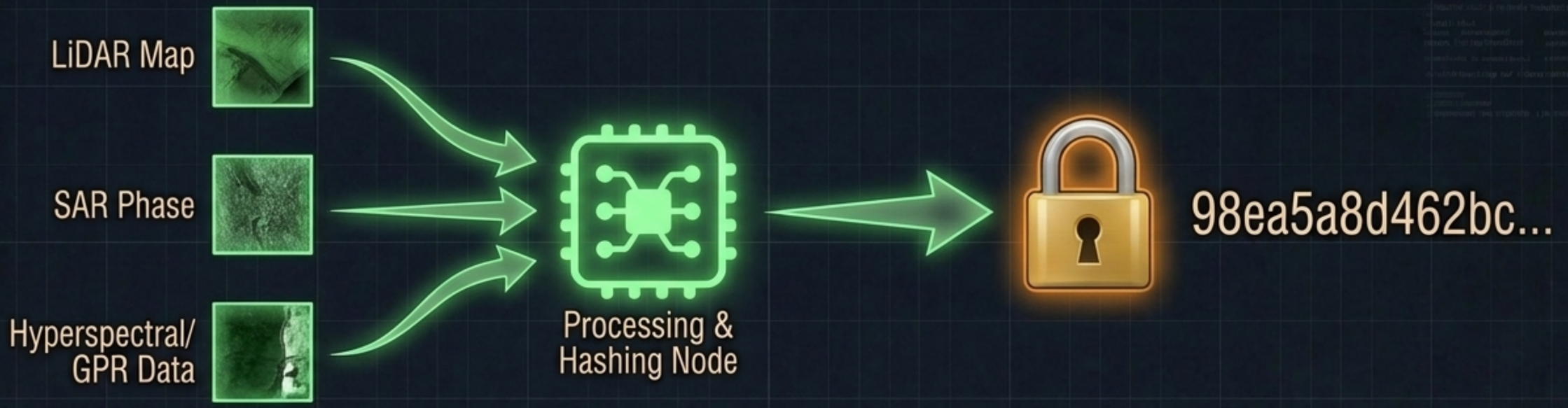
The Evidence Dependency Graph



Cryptographic Empirical Physics:
Algorithms that find overlooked anomalies in old data must be mathematically defensible.

The Ledger:
Every ingested LiDAR map, SAR phase, and spatial overlay is hashed via SHA-256 upon acquisition.

The Proof:
This creates an immutable, timestamped audit history. It proves to LE that the physical anomaly existed on the Lindeman property before the search began, securing chain of custody.

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Synthesis: The 4-Zone Target Grid

1. **Zone 1: The Primary Anchor (Deterministic / 95%+)**
   - Immediate Excavation. Fixed structural anomalies where LiDAR trench geometry intersects directly with brush-clearing data.

2. **Zone 2: The Catchment Funnel (Hydraulic / 80%+)**
   - Coring required. Gravity flow channels leading into the pond where evidence naturally aggregates.

3. **Zone 3: The Proxy Anomaly (Predictive / 50%+)**
   - Grid sweep area. Intersecting signals of SIPI canopy stress and potential cellular ping cutoffs.

4. **Zone 4: The Speculative Fringe (Excluded / <50%)**
   - Automated exclusion. Unverified hearsay that contradicts the terrain physics. Do not deploy assets here.

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Actionable Deployment Protocol

Step 1: > RETRIEVE ARCHIVAL DATA

Pull historic bare-earth LiDAR, September 2026 satellite imagery, and localized GIS parcel data for 469 Dogwood Ln.

Step 2: > RUN SPATIAL OVERLAY

Execute the HFMR engine. Cross-reference the 9-day reporting gap against localized Nitrogen flushes (SIPI) and topographic micro-depressions.

Step 3: > SECURE IMMUTABLE HASH

Generate a SHA-256 ledger of the Zone 1 Target Coordinates to establish an unassailable evidentiary timeline for the McDonald County Sheriff's Office.

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